[1]

スカラー：任意の変換により不変な量

直交座標系 Σ, Σ' に関するテンソルをそれぞれ T_{ij}, T'_{ij}

Σ から Σ' への座標変換を a_{ij} とする．$\forall x_i, y_i$ について

$$\begin{cases} x_j' = a_{jk} x_k \\ y_i' = a_{ij} y_j \end{cases} \qquad \begin{cases} y_j = T_{jk} x_k \\ y_i' = T'_{ij} x_j' \end{cases}$$

$$\rightarrow \quad a_{ij} T_{jk} = T'_{ij} a_{jk} \qquad \therefore |T_{ij}| = |T'_{ij}|$$

[2]

直交基底 $\mathbf{e}_k$ $\quad \mathbf{a} = a_k \mathbf{e}_k$, etc

$(\mathbf{e}_i \times \mathbf{e}_j)_k = \epsilon_{ijk}$ より

$(\mathbf{a} \times \mathbf{b})_k = a_i b_j \epsilon_{ijk}$

$(\nabla \times \mathbf{v})_k = \partial_i v_j \epsilon_{ijk}$

$\nabla \cdot \mathbf{v} = \partial_i v_i$

$(\nabla \phi)_k = \partial_k \phi$

$\Delta \phi = \partial_i \partial_i \phi$

[3]

$$\epsilon_{ijk} = (\mathbf{e}_i \times \mathbf{e}_j)_k = (\mathbf{e}_i \times \mathbf{e}_j) \cdot \mathbf{e}_k = \det(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k)$$

(1) $$\epsilon_{ijk}\epsilon_{\ell mk} = \sum_k \det{}^t(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k) \cdot (\mathbf{e}_\ell, \mathbf{e}_m, \mathbf{e}_k)$$

$$= \sum_k \begin{vmatrix} \delta_{i\ell} & \delta_{im} & \delta_{ik} \\ \delta_{j\ell} & \delta_{jm} & \delta_{jk} \\ \delta_{k\ell} & \delta_{km} & \delta_{kk} \end{vmatrix}$$

$$= \sum_k \delta_{i\ell}\delta_{jm}\delta_{kk} + \delta_{im}\delta_{jk}\delta_{\ell k} + \delta_{ik}\delta_{j\ell}\delta_{mk} \qquad (\delta_{kk} = 3,\ \delta_{ik}\delta_{mk} = \delta_{im})$$
$$- \delta_{ik}\delta_{jm}\delta_{k\ell} - \delta_{i\ell}\delta_{jk}\delta_{mk} - \delta_{im}\delta_{j\ell}\delta_{kk}$$

$$= 3(\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}) + 2(\delta_{im}\delta_{j\ell} - \delta_{i\ell}\delta_{jm})$$

$$= \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}$$

4階テンソル $= a\,\delta_{ij}\delta_{\ell m} + b\,\delta_{i\ell}\delta_{jm} + c\,\delta_{im}\delta_{j\ell}$

$a = 0,\ b = 1,\ c = -1$

(2) $$\epsilon_{ijk}\epsilon_{\ell jk} = \sum_j \delta_{i\ell}\delta_{jj} - \delta_{ij}\delta_{j\ell} = 2\delta_{i\ell}$$

2階テンソル $= c\,\delta_{i\ell}$, $c = 2$

(3) $$\epsilon_{ijk}\epsilon_{ijk} = 6$$

[4]

(1) $\{\nabla\times(\nabla\times A)\}_k = \partial_i (\nabla\times A)_j \epsilon_{ijk} = \partial_i \partial_\ell A_m \epsilon_{ijk}\epsilon_{\ell mj}$

$= \partial_i \partial_\ell A_m (\delta_{im}\delta_{k\ell} - \delta_{i\ell}\delta_{km})$

$= \partial_k(\partial_m A_m) - (\partial_\ell \partial_\ell) A_k = \{\nabla(\nabla\cdot A)\}_k - \{\nabla^2 A\}_k$

(2) $\{(A\cdot\nabla)B + (B\cdot\nabla)A + A\times(\nabla\times B) + B\times(\nabla\times A)\}_k$

$= (A_i\partial_i)B_k + (B_i\partial_i)A_k + A_i\partial_\ell B_m \epsilon_{ijk}\epsilon_{\ell mj} + B_i\partial_\ell A_m \epsilon_{ijk}\epsilon_{\ell mj}$

$= A_i\partial_i B_k + B_i\partial_i A_k + (A_i\partial_k B_i - A_i\partial_i B_k) + (B_i\partial_k A_i - B_i\partial_i A_k)$

$= \partial_k(A_i B_i) = \{\nabla(A\cdot B)\}_k$

(3) $\nabla\cdot(A\times B) = \partial_k (A\times B)_k = \partial_k A_i B_j \epsilon_{ijk}$

$= B_j(\partial_k A_i \epsilon_{ijk}) - A_i(\partial_k B_j \epsilon_{kji})$

$= B\cdot(\nabla\times A) - A\cdot(\nabla\times B)$

(4) $\{\nabla\times(A\times B)\}_k = \partial_i (A\times B)_j \epsilon_{ijk} = \partial_i A_\ell B_m \epsilon_{ijk}\epsilon_{\ell mj}$

$= \partial_i A_\ell B_m (\delta_{im}\delta_{k\ell} - \delta_{i\ell}\delta_{km})$

$= \partial_i A_k B_i - \partial_i A_i B_k$

$= A_k(\partial_i B_i) + (B_i\partial_i)A_k - (A_i\partial_i)B_k - B_k(\partial_i A_i)$

$= \{A(\nabla\cdot B) + (B\cdot\nabla)A - (A\cdot\nabla)B - B(\nabla\cdot A)\}_k$

[5]

(1) $\nabla r = \hat{r}$ (2) $\nabla \cdot \vec{r} = 3$ (3) $\nabla \times \vec{r} = 0$

(4) $\nabla f(r) = \frac{\partial f}{\partial r} \nabla r = f'(r)\, \hat{r}$

(5) $\nabla \cdot (f(r)\vec{r}) = \{\nabla f(r)\} \cdot \vec{r} + f(r)(\nabla \cdot \vec{r}) = f'(r)\, r + 3 f(r)$

(6) $\nabla \times (f(r)\vec{r}) = f(r)(\nabla \times \vec{r}) - \vec{r} \times (\nabla f(r)) = 0$

(7) $\nabla^2 \left(\frac{1}{r}\right) = \nabla \cdot \nabla \left(\frac{1}{r}\right) = \nabla \cdot \left(-\frac{1}{r^3}\vec{r}\right) = 0$

(8) $\nabla \cdot (\vec{A} \times (\vec{r} \times \vec{A})) = \partial_k A_i\, x_l A_m\, \epsilon_{ijk}\, \epsilon_{lmj}$

$= \partial_k A_i\, x_l A_m (\delta_{im}\delta_{kl} - \delta_{il}\delta_{km})$

$= \partial_k A_i\, x_k A_i - \partial_k A_i\, x_i A_k$

$= 3|\vec{A}|^2 - |\vec{A}|^2 = 2|\vec{A}|^2$

(9) $\{\nabla \times ((\vec{A} \cdot \vec{r})\vec{B})\}_k = \partial_i A_l\, x_l B_j\, \epsilon_{ijk} = A_i B_j \epsilon_{ijk} = \{A \times B\}_k$

(10) $\{\nabla (\vec{A} \cdot (\vec{B} \times \vec{r}))\}_k = \partial_l A_i B_j\, x_k\, \epsilon_{jki} = A_i B_j \epsilon_{jki} = \{A \times B\}_k$

[6]

$\{\nabla \times (\nabla f)\}_k = \partial_i \partial_j f\, \epsilon_{ijk} = \partial_i^2 f\, \epsilon_{iik} = 0$

$\nabla \cdot (\nabla \times A) = \partial_k \partial_i A_j\, \epsilon_{ijk} = \partial_i^2 A_j\, \epsilon_{iji} = 0$

[7]

(1) 発散定理 $\int_V \mathrm{div}\,\vec{A}\,dv = \int_S \vec{A}\cdot\vec{n}\,dS$ より

$$\int_S \vec{r}\cdot\vec{n}\,dS = \int_V \mathrm{div}\,\vec{r}\,dv = 3V$$

(2) $$\int_S (\mathrm{rot}\,\vec{A})\cdot\vec{n}\,dS = \int_V \mathrm{div}\,(\mathrm{rot}\,\vec{A})\,dv = 0$$

[8]

(1) $dx_i = J_{ij}\, d\xi^j \quad \left(J_{ij} = \dfrac{\partial x_i}{\partial \xi_j} \right)$

$$\delta_{ij}\, dx_i\, dx_j = \delta_{ij} (J_{ik}\, d\xi^k)(J_{jl}\, d\xi^l) = J_{ik} J_{il}\, d\xi^k d\xi^l = g_{kl}\, d\xi^k d\xi^l$$

$$\int (\nabla\phi)(\nabla\chi)\, d^3r = \int \left(\frac{\partial \xi^i}{\partial x_k} \cdot \frac{\partial \phi}{\partial \xi^i} \right) \cdot \left(\frac{\partial \xi^j}{\partial x_k} \cdot \frac{\partial \chi}{\partial \xi^j} \right) d^3r \qquad \nabla_i = \frac{\partial \xi^i}{\partial x_i} \cdot \frac{\partial}{\partial \xi^i}$$

$$= \int \underbrace{J^{ki} J^{kj}}_{\| \; g^{ij}} \frac{\partial \phi}{\partial \xi^i} \cdot \frac{\partial \chi}{\partial \xi^j} \cdot \underbrace{(\det J_{ij})}_{\| \; \sqrt{\det g_{ij}}} \cdot d^3\xi$$

(2) $$\int \left(g^{ij} \frac{\partial \phi}{\partial \xi^i} \sqrt{g} \right) \frac{\partial \chi}{\partial \xi^j}\, d^3\xi = \underbrace{\left[\chi \left(g^{ij} \frac{\partial \phi}{\partial \xi^i} \sqrt{g} \right) \right]_{-\infty}^{\infty}}_{\| \; 0} - \int \chi \frac{\partial}{\partial \xi^j} \left(g^{ij} \frac{\partial \phi}{\partial \xi^i} \sqrt{g} \right) d^3\xi$$

$$\therefore \nabla^2\phi = \frac{1}{\sqrt{g}} \frac{\partial}{\partial \xi^j} \left(g^{ij} \frac{\partial \phi}{\partial \xi^i} \sqrt{g} \right)$$

(3) 極座標

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2\theta \end{pmatrix}$$

円柱座標

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

[9]

円柱座標　$\Phi = P(\rho)\Theta(\varphi)Z(z)$

$$\frac{1}{P}\left(\frac{d^2P}{d\rho^2}+\frac{1}{\rho}\frac{dP}{d\rho}\right)+\frac{1}{\rho^2\Theta}\frac{d^2\Theta}{d\varphi^2}+\frac{1}{Z}\frac{d^2Z}{dz^2}+k^2=0 \qquad (1)$$

$$\begin{cases} \dfrac{d^2Z}{dz^2}-\alpha Z=0 & (2) \\ \dfrac{\rho^2}{P}\left(\dfrac{d^2P}{d\rho^2}+\dfrac{1}{\rho}\dfrac{dP}{d\rho}\right)+(k^2+\alpha)\rho^2+\dfrac{1}{\Theta}\dfrac{d^2\Theta}{d\varphi^2}=0 & (3) \end{cases}$$

(3) → $\dfrac{d^2\Theta}{d\varphi^2}-\beta\Theta=0$, $\dfrac{d^2P}{d\rho^2}+\dfrac{1}{\rho}\dfrac{dP}{d\rho}+\left(k^2+\alpha+\dfrac{\beta}{\rho^2}\right)P=0$ (4)

Θ は1価関数であるから $\beta=-n^2$, さらに $\sqrt{k^2+\alpha}\,\rho=x$ とおくと

(4) → $\dfrac{d^2P}{dx^2}+\dfrac{1}{x}\dfrac{dP}{dx}+\left(1-\dfrac{n^2}{x^2}\right)P=0$ (Bessel の微分方程式)

一般解　$A J_n(x)+B J_{-n}(x)$

$$J_n(x)=\sum_k \frac{(-1)^k}{k!\,\Gamma(n+k+1)}\left(\frac{x}{2}\right)^{n+2k} \qquad \Gamma(x)=\int_0^\infty e^{-t}t^{x-1}dt$$

極座標、 $\phi = R(r)Y(\theta,\varphi)$

$$\frac{r^2}{R}\left(\frac{d^2R}{dr^2}+\frac{2}{r}\frac{dR}{dr}+k^2R\right)+\frac{1}{Y\sin\theta}\cdot\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{Y\sin^2\theta}\cdot\frac{\partial^2Y}{\partial\varphi^2}=0 \qquad (1)$$

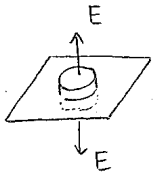
$$\begin{cases} \dfrac{d^2R}{dr^2}+\dfrac{2}{r}\dfrac{dR}{dr}+\left(k^2-\dfrac{\mu}{r^2}\right)R=0 & (2) \\ \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial Y}{\partial\theta}\right)+\dfrac{1}{\sin^2\theta}\dfrac{\partial^2Y}{\partial\varphi^2}+\mu Y=0 & (3) \end{cases}$$

いまz軸が対称軸となっている場合を考える. $\cos\theta = x$ とおくと

(3) → $\dfrac{d}{dx}\left\{(1-x^2)\dfrac{dY}{dx}\right\}+m(m+1)Y=0$

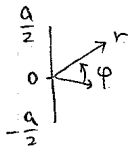
[10]

(1)



$\epsilon E \cdot 2S = \sigma S \rightarrow E = \frac{\sigma}{2\epsilon}$ $\quad \phi = -\frac{\sigma}{2\epsilon}|\vec{r}\cdot\vec{n}|$

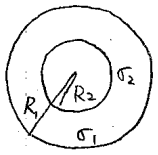
(2)



$$\phi = \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{\lambda dz'}{\sqrt{r^2+(z'-z)^2}} = \lambda \int_{-\frac{a}{2}-z}^{\frac{a}{2}-z} \frac{dz}{\sqrt{r^2+z^2}} = \lambda \ln \frac{\sqrt{r^2+\left(\frac{a}{2}-z\right)^2}+\frac{a}{2}-z}{\sqrt{r^2+\left(-\frac{a}{2}-z\right)^2}-\frac{a}{2}-z}$$

E:困各

(3)



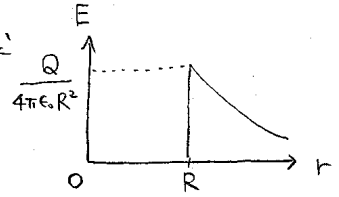
$0 < r < R_2$; $E = 0$

$R_2 < r < R_1$; $E = \frac{\sigma_2 R_2^2}{\epsilon r^2}$

$R_1 < r$; $E = \frac{\sigma_1 R_1^2 + \sigma_2 R_2^2}{\epsilon r^2}$

[11]

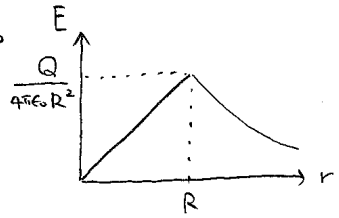
導体球



$0 < r < R$; $E = 0$

$R < r$; $E = \frac{Q}{4\pi\epsilon_0 r^2}$

一様分布

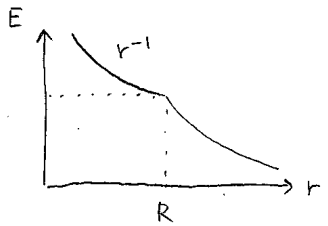


$0 < r < R$; $E = \frac{Q}{4\pi\epsilon_0} \cdot \frac{r}{R^3}$

$R < r$; $E = \frac{Q}{4\pi\epsilon_0 r^2}$

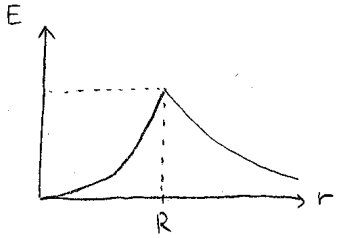
球対称

$n = -2$



$0 < r < R$; $E = \frac{Q}{4\pi\epsilon_0 R^2} \cdot \left(\frac{r}{R}\right)^{n+1}$ $\quad (n > -3)$

$n = 2$



[12]

$$\phi(r) = \frac{q}{4\pi\epsilon_0 a}\left(1+\frac{a}{r}\right)e^{-\frac{2r}{a}}$$

$$E_r = -\frac{\partial\phi}{\partial r} = \frac{q}{4\pi\epsilon_0 a}\left(\frac{2}{a}+\frac{2}{r}+\frac{a}{r^2}\right)e^{-\frac{2r}{a}}$$

$$\rho = \epsilon_0 \nabla\cdot\vec{E} = \epsilon_0 \frac{1}{r^2}\frac{d}{dr}(r^2 E_r) = -\frac{q}{\pi a^3}e^{-\frac{2r}{a}} + q\,\delta(r)$$

[13]

$$\nabla^2\phi(r,\theta,\varphi) = 0 \quad \rightarrow \quad \phi = \sum_{l=0}^{\infty}\sum_{m=-l}^{l}\left[A_{lm}r^l + B_{lm}r^{-l-1}\right]P_l^{(m)}(\cos\theta)\,e^{im\varphi}$$

Boundary $\begin{cases} \phi(R,\theta) = V \\ \phi(r,\theta) \xrightarrow[r\to\infty]{} -E_0 r\cos\theta \end{cases}$

$$\phi = \left(A_0 + \frac{B_0}{r}\right) + E_0\left(-r + \frac{R^3}{r^2}\right)\cos\theta$$

球面上の電荷密度

$$\sigma = \epsilon_0 E_r = -\epsilon_0 \frac{\partial\phi}{\partial r}\Big|_{r=R} = \epsilon_0\frac{B_0}{R^2} + 3\epsilon_0 E_0\cos\theta$$

全電荷は

$$Q = \int_0^{\pi}\sigma(\theta)\cdot(2\pi R\sin\theta)\cdot R\,d\theta = 4\pi\epsilon_0 B_0$$

$$\therefore\ \phi = V - \frac{Q}{4\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 r} - E_0 r\cos\theta + \frac{E_0 R^3}{r^2}\cos\theta$$

[14]

$$\frac{1}{2}\sum_i q_i\phi_i = \frac{1}{2}\int \rho\phi\, dV = \frac{1}{2}\epsilon_0\int(\nabla\phi)\cdot(\nabla\phi)\, dV - \frac{1}{2}\epsilon_0\int \nabla\cdot(\phi\nabla\phi)\, dV$$

$$= \frac{1}{2}\epsilon_0\int E^2 dV + \frac{1}{2}\epsilon_0\int \phi\vec{E}\cdot d\vec{S}$$

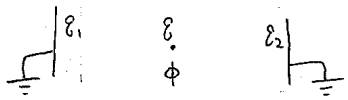
$\frac{1}{r}\cdot\frac{1}{r^2}\cdot r^2 \to 0$

[15] グリーンの定理

$$\sum_i (q_i\phi_i' - q_i'\phi_i) = \int_V (\rho\phi' - \rho'\phi)\, dV = \epsilon_0\int_V (\phi\cdot\nabla^2\phi' - \phi'\cdot\nabla^2\phi)\, dV$$

$$= \epsilon_0\int_S (\phi\cdot\nabla\phi' - \phi'\nabla\phi)\cdot\vec{n}\, dS$$

$$= \epsilon_0\int_S (\phi\vec{E}' - \phi'\vec{E})\cdot\vec{n}\, dS = 0$$

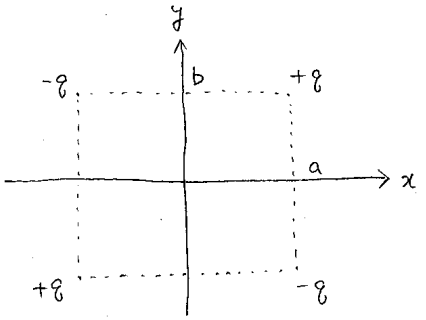


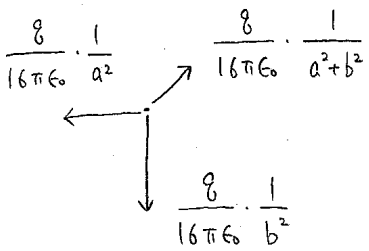
[16] 電気映像法

$$\Phi = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R_{--}} + \frac{1}{R_{++}} - \frac{1}{R_{+-}} - \frac{1}{R_{-+}}\right)$$

$$R_{\pm\pm} = \sqrt{(x\pm a)^2 + (y\pm b)^2 + z^2}$$

$$\Phi(x=0) = \Phi(y=0) = 0$$





[17]

$$V = e\phi\left(\vec{r}+\frac{\vec{s}}{2}\right) - e\phi\left(\vec{r}-\frac{\vec{s}}{2}\right) \simeq e\left\{\phi(\vec{r}) + \frac{1}{2}\vec{s}\cdot\nabla\phi(\vec{r})\right\} - e\left\{\phi(\vec{r}) - \frac{1}{2}\vec{s}\cdot\nabla\phi(\vec{r})\right\}$$

$$= e\vec{s}\cdot\nabla\phi(\vec{r}) = -\vec{p}\cdot\vec{E}(\vec{r})$$

$$\therefore \vec{F} = \nabla(\vec{p}\cdot\vec{E}) = (\vec{p}\cdot\nabla)\vec{E} + (\vec{E}\cdot\nabla)\vec{p} + \vec{p}\times(\nabla\times\vec{E}) + \vec{E}\times(\nabla\times\vec{p})$$

$$(\vec{E}\cdot\nabla)\vec{p} = 0,\quad \vec{p}\times(\nabla\times\vec{E}) = 0,\quad \vec{E}\times(\nabla\times\vec{p}) = 0$$

$$\phi = \frac{e}{4\pi\epsilon_0}\left(\frac{1}{\left|\vec{r}+\frac{\vec{s}}{2}\right|} - \frac{1}{\left|\vec{r}-\frac{\vec{s}}{2}\right|}\right) \simeq \frac{e}{4\pi\epsilon_0}\vec{s}\cdot\nabla\left(\frac{1}{|\vec{r}|}\right) = \frac{\vec{p}\cdot\vec{r}}{4\pi\epsilon_0 r^3}$$

$$\vec{E} = -\nabla_r\phi = -\frac{1}{4\pi\epsilon_0}\left[(\vec{p}\cdot\vec{r})\nabla_r\left(\frac{1}{r^3}\right) + \frac{1}{r^3}\nabla_r(\vec{p}\cdot\vec{r})\right]$$

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{3(\vec{p}\cdot\vec{r})\vec{r} - r^2\vec{p}}{r^5}\right] - \frac{1}{4\pi\epsilon_0}\cdot\frac{4\pi}{3}\vec{p}\cdot\delta(\vec{r})\ ?$$

相互作用エネルギー

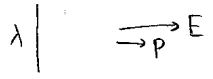
$$U = -\vec{p}_1\cdot\frac{1}{4\pi\epsilon_0}\cdot\frac{3(\vec{p}_2\cdot\vec{r})\vec{r} - r^2\vec{p}_2}{r^5} = \frac{1}{4\pi\epsilon_0}\left[-\frac{3}{r^5}(\vec{p}_1\cdot\vec{r})(\vec{p}_2\cdot\vec{r}) + \frac{1}{r^3}(\vec{p}_1\cdot\vec{p}_2)\right]$$

$$\vec{F} = -\nabla_r U = \frac{1}{4\pi\epsilon_0}\left[3(\vec{p}_1\cdot\vec{r})(\vec{p}_2\cdot\vec{r})\nabla_r\left(\frac{1}{r^5}\right) + \frac{3}{r^5}\nabla_r\left\{(\vec{p}_1\cdot\vec{r})(\vec{p}_2\cdot\vec{r})\right\} - (\vec{p}_1\cdot\vec{p}_2)\nabla_r\left(\frac{1}{r^3}\right)\right]$$

$$= \frac{1}{4\pi\epsilon_0}\left[-\frac{15(\vec{p}_1\cdot\vec{r})(\vec{p}_2\cdot\vec{r})\vec{r}}{r^7} + \frac{3\left\{(\vec{p}_2\cdot\vec{r})\vec{p}_1 + (\vec{p}_1\cdot\vec{r})\vec{p}_2 + (\vec{p}_1\cdot\vec{p}_2)\vec{r}\right\}}{r^5}\right]$$

[18]

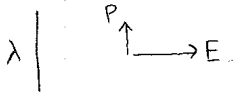
(1) parallel



トルク $\vec{N} = \vec{p}\times\vec{E} = 0$

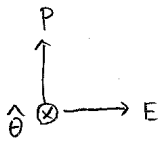
力 $\vec{F} = \nabla(\vec{p}\cdot\vec{E}) = -\dfrac{\lambda p\vec{r}}{2\pi\epsilon_0 r^3}$

(2) perpendicular

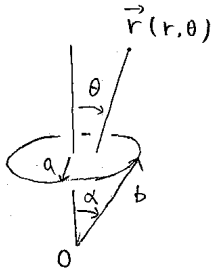


トルク $\vec{N} = \vec{p}\times\vec{E} = \dfrac{\lambda p}{2\pi\epsilon_0 r}\hat{\theta}$

力 $\vec{F} = \nabla(\vec{p}\cdot\vec{E}) = 0$

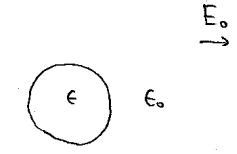


[19]



[20]

Boundary $\quad \epsilon \frac{\partial \phi_{in}}{\partial r} = \epsilon \frac{\partial \phi_{ext}}{\partial r}$, $\frac{\partial \phi_{in}}{\partial \theta} = \frac{\partial \phi_{ext}}{\partial \theta}$



$$\phi_{in}(r,\theta) = A_1 r\cos\theta = -\frac{3\epsilon_0}{\epsilon+2\epsilon_0} E_0 r\cos\theta$$

$$\phi_{ext}(r,\theta) = \left(Ar + \frac{B}{r^2}\right)\cos\theta = -E_0 r\cos\theta + \frac{\epsilon-\epsilon_0}{\epsilon+2\epsilon_0}\cdot\frac{a^3}{r^2} E_0\cos\theta$$

[21]

$$\phi_{in} = Ar\mu + \cdots$$

$$\phi = \frac{B}{r} + \left(Cr + \frac{D}{r^2}\right)\mu + \cdots$$

$$\phi_{ext} = \frac{F}{r} + \left(-E_0 + \frac{G}{r^2}\right)\mu \quad \therefore \phi_{ext} \to -E_0 r\mu$$

Boundary

$\frac{\partial}{\partial\mu}$ $\quad AR_1 = CR_1 + \frac{D}{R_1}$, $CR_2 + \frac{D}{R_2^2} = -E_0 R_2 + \frac{G}{R_2^2}$

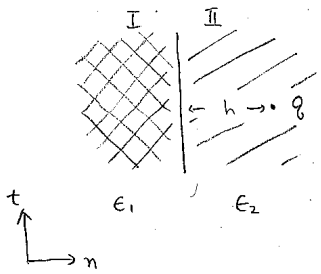
$\epsilon\frac{\partial}{\partial r}$ $\quad \epsilon_0 A\mu = -\epsilon\frac{B}{R_1^2} + \epsilon\left(C - \frac{2D}{R_1^3}\right)\mu$, $-\epsilon\frac{B}{R_2^2} + \epsilon\left(C - \frac{2D}{R_2^3}\right)\mu = -\epsilon_0\frac{F}{R_2^2} + \epsilon_0\left(-E_0 - \frac{2G}{R_2^3}\right)\mu$

$$\to \begin{cases} A = C + \dfrac{D}{R_1^3} \ , \quad C + \dfrac{D}{R_2^3} = -E_0 + \dfrac{G}{R_2^3} \\ B = 0 \ , \quad \epsilon_0 A = \epsilon\left(C - \dfrac{2D}{R_1^3}\right) \ , \quad F = 0 \ , \quad \epsilon\left(C - \dfrac{2D}{R_2^3}\right) = \epsilon_0\left(-E_0 - \dfrac{2G}{R_2^3}\right) \end{cases}$$

$$D = \frac{3\epsilon_0(\epsilon_0-\epsilon)R_1^3 R_2^3}{\{(2\epsilon_0-\epsilon)(\epsilon_0+2\epsilon)R_2^3 - 2(\epsilon_0-\epsilon)(\epsilon_0+\epsilon)R_1^3\}} E_0$$

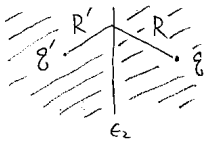
$$C = \frac{-3\epsilon_0(\epsilon_0+2\epsilon)R_2^3}{\{\quad\quad\quad

[22] 誘電体の映像法



Boundary $\quad \epsilon_1 \frac{\partial \phi_1}{\partial n} = \epsilon_2 \frac{\partial \phi_2}{\partial n}, \quad \phi_1 = \phi_2$

$$\phi_2 = \frac{1}{4\pi\epsilon_2} \cdot \left(\frac{q}{R} + \frac{q'}{R'} \right)$$

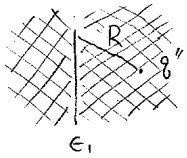


$$R = \sqrt{(h-n)^2 + t^2}$$

$$R' = \sqrt{(h+n)^2 + t^2}$$

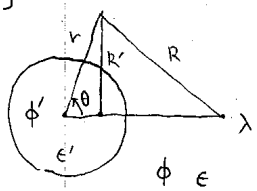
$$\phi_1 = \frac{1}{4\pi\epsilon_1} \cdot \frac{q''}{R}$$

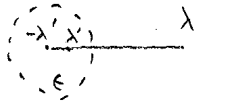
q のまわりの誘電を含めて q'' とする



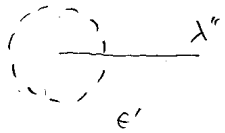
$$\frac{\partial}{\partial n}\left(\frac{1}{R}\right)_{n=0} = \frac{2h}{R^3} \quad \text{etc}$$

[23]



φ 

$$\phi = -\frac{1}{2\pi\epsilon}(\lambda \ln R + \lambda' \ln R' - \lambda' \ln r)$$

φ' 

$$\phi' = -\frac{1}{2\pi\epsilon'}\lambda'' \ln R$$

$$R^2 = b^2 + r^2 - 2br\cos\theta$$

$$R'^2 = \left(\frac{a^2}{b}\right)^2 + r^2 - 2\frac{a^2}{b} r\cos\theta$$

Boundary ($\forall_\mu$)

$$\frac{\partial}{\partial\mu}\Big|_a \quad \frac{\lambda}{\epsilon}\cdot\frac{ab}{R^2} + \frac{\lambda'}{\epsilon}\cdot\frac{a^2/b}{R'^2} = \frac{\lambda''}{\epsilon'}\cdot\frac{ab}{R^2} \rightarrow \underbrace{\left(\frac{\lambda}{\epsilon} + \frac{\lambda'}{\epsilon} - \frac{\lambda''}{\epsilon'}\right)}_{①} \cdot \frac{ab}{a^2+b^2-2ab\mu} = 0$$

$$\epsilon\frac{\partial}{\partial r}\Big|_a \quad \lambda\frac{a-b\mu}{R^2} + \lambda'\frac{a-\frac{a^2}{b}\mu}{R'^2} - \frac{\lambda'}{a} = \lambda''\frac{a-b\mu}{R^2}$$

$R'^2 = \frac{a^2}{b^2}R^2$

$$\rightarrow (\lambda-\lambda'')(a^2-ab\mu) + \lambda'(b^2-ab\mu) = \lambda' R^2$$

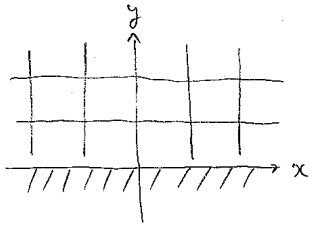
$$\rightarrow \underbrace{(\lambda-\lambda''-\lambda')}_{②} a^2 - ab\mu\underbrace{(\lambda-\lambda''-\lambda')} = 0$$

$$\begin{cases} \lambda' = \dfrac{\epsilon-\epsilon'}{\epsilon+\epsilon'}\lambda \\ \lambda'' = \dfrac{2\epsilon'}{\epsilon+\epsilon'}\lambda \end{cases}$$

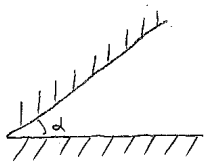
単位長さあたりに働く力

$$f = \frac{1}{2\pi\epsilon}\left(\frac{\lambda'}{b-\frac{a^2}{b}} - \frac{\lambda'}{b}\right)$$

[24]



analystic →



ω

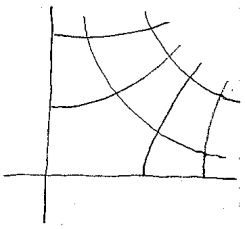
$\omega = z^{\frac{\alpha}{\pi}}$

($z \neq 0$ 2つ等角)

電気力線　$x = a \rightarrow \omega = (a + iy)^{\frac{\alpha}{\pi}}$

等電位面　$y = b \rightarrow \omega = (x + ib)^{\frac{\alpha}{\pi}}$

ex) $\alpha = \frac{\pi}{2}$　$a + ib = r^2 e^{2i\theta} \rightarrow \begin{cases} a = r^2(x^2 - y^2) \\ b = 2r^2 xy \end{cases}$

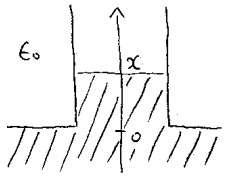


$\boxed{\Delta\phi = 0}$　正則関数 $f(z) = f(x+iy) = u(x,y) + iv(x,y)$

Cauchy - Riemann condition　$\Delta u = \Delta v = 0$

[25]

[26]



ϵ, ρ

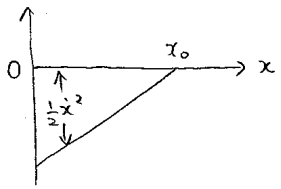
$$C = \epsilon_0 \frac{a(a-x)}{d} + \epsilon \frac{ax}{d} = \frac{a}{d}\left(\epsilon_0 a + (\epsilon - \epsilon_0)x\right)$$

$$\begin{cases} F_x\,dx + dU = V\,dQ \\ d\left(\frac{1}{2}M\dot{x}^2\right) + d\left(\frac{1}{2}Mgx\right) = F_x\,dx \qquad (M = \rho x a d) \end{cases}$$

$$\rightarrow \quad \frac{1}{2}M\dot{x}^2 + \frac{1}{2}Mgx - \frac{1}{2}CV^2 = \text{const}$$

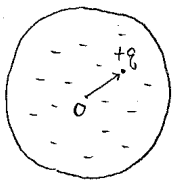
$$\frac{1}{2}x\dot{x}^2 + \frac{1}{2}gx^2 - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2}x = 0 \qquad (t=0;\ x=\dot{x}=0)$$

$$\therefore \quad \frac{1}{2}\dot{x}^2 + \frac{1}{2}gx - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2} = 0$$



$$x_0 = \frac{(\epsilon-\epsilon_0)V^2}{\rho g d^2} \quad ?$$

[27]

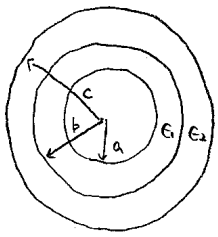


$$E_- = \frac{q}{4\pi\epsilon_0}\cdot\frac{r}{a^3}$$

$$P = nqr$$

$$\epsilon = \epsilon_0 + \frac{P}{E} = \epsilon_0(1 + 4\pi a^3 n)$$

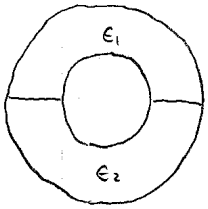
[28]



$E = \frac{Q}{4\pi\epsilon r^2}$ より

$$V = \frac{Q}{4\pi\epsilon_1}\int_a^b \frac{dr}{r^2} + \frac{Q}{4\pi\epsilon_2}\int_b^c \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_1}\left(\frac{1}{a}-\frac{1}{b}\right) + \frac{Q}{4\pi\epsilon_2}\left(\frac{1}{b}-\frac{1}{c}\right)$$

$$\therefore\ C = \frac{Q}{V} = \left(\frac{\frac{1}{a}-\frac{1}{b}}{4\pi\epsilon_1} + \frac{\frac{1}{b}-\frac{1}{c}}{4\pi\epsilon_2}\right)^{-1}$$ 」



球対称と仮定

$$2\pi r^2(\epsilon_1+\epsilon_2)E = Q$$

$$V = \frac{Q}{2\pi(\epsilon_1+\epsilon_2)}\int_a^c \frac{dr}{r^2} = \frac{Q}{2\pi(\epsilon_1+\epsilon_2)}\cdot\left(\frac{1}{a}-\frac{1}{c}\right)$$

$$\therefore\ C = \frac{Q}{V} = \frac{2\pi(\epsilon_1+\epsilon_2)}{\frac{1}{a}-\frac{1}{c}}$$ 」

ラプラス方程式をみたす

唯一つの解だから O.K.

$$\begin{cases} \nabla \cdot E = \frac{1}{\epsilon_0}\rho & E = -\nabla\phi - \frac{\partial A}{\partial t} \\ \nabla \times E = -\frac{\partial B}{\partial t} & \\ \nabla \cdot B = 0 & B = \nabla \times A \\ \frac{1}{\mu_0}\nabla \times B = J + \epsilon_0 \frac{\partial E}{\partial t} & \end{cases}$$

Lorentz gauge $\quad \nabla \cdot A + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0$

$$\Box = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2$$

$$\begin{cases} \Box \phi = \frac{1}{\epsilon_0}\rho \\ \Box A = \mu_0 J \end{cases}$$

$$\nabla^2 A = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 A}{\partial \phi^2} + \frac{\partial^2 A}{\partial z^2} = -\mu_0 J\, \delta(\rho)$$

$$A_\phi = A_z = 0$$

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) = 0 \quad \rightarrow \quad A_z = c \ln \rho + c_0 \qquad ?$$

[30]

$$M = \frac{1}{2}\int \vec{r} \times \vec{j}\, dV = \frac{1}{2} r e v = \frac{e}{2m} L \qquad (L = mrv)$$



$$E \cdot 2\pi r = -\pi r^2 \frac{dB}{dt}$$

$$E = -\frac{1}{2} r \frac{dB}{dt}$$

$$m\frac{dv}{dt} = -eE = \frac{1}{2} er \frac{dB}{dt} \quad \rightarrow \quad \Delta v = \frac{er}{2m} \Delta B$$

$$\therefore \Delta M = \frac{1}{2} re \Delta v = \frac{e^2 r^2}{4m} \Delta B$$

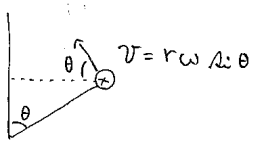
$$\frac{d\vec{L}}{dt} = \vec{M} \times \vec{B} \quad \rightarrow \quad \frac{e}{2m} \cdot \frac{d\vec{M}}{dt} = \vec{M} \times \vec{B}$$

[31]

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{d\vec{I}\times\vec{r}}{R^2} \qquad B(z) = \frac{\mu_0}{2}\left(\frac{I_A}{a}\cdot\left(1+\frac{z^2}{a^2}\right)^{-3/2} - \frac{I_B}{b}\left(1+\frac{z^2}{b^2}\right)^{-3/2}\right)$$

$$\frac{I_A}{a^3} = \frac{I_B}{b^3}$$

[32] $\vec{M} = \frac{1}{2}\int \vec{r}\times\vec{j}(\vec{r})\,d\vec{V}$



$$M_f = \frac{1}{2}\int r\cdot\rho\, r\omega\sin\theta\cdot\sin\theta\cdot(2\pi r\sin\theta)\cdot r\,dr\,d\theta$$

$$= \pi\rho\omega\int r^4 dr\int \sin^3\theta\,d\theta \qquad \int_{-1}^{1}(\mu^2-1)\,d\mu = \frac{4}{3}$$

$$= \pi\rho\omega\,\frac{R^5}{5}\cdot\frac{4}{3} = \frac{4}{15}\pi\rho\omega R^5$$

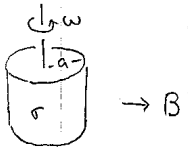
$$M_s = \frac{1}{2}\int R\cdot\sigma R\omega\sin\theta\cdot\sin\theta\cdot(2\pi R\sin\theta)\,R\,d\theta$$

$$= \pi\sigma\omega R^4\int\sin^3\theta\,d\theta = \frac{4}{3}\pi\sigma\omega R^4,$$

[33] $LI = \Phi = nl\,B\cdot S \qquad B = \mu_0 nI,\ S = \pi a^2$

$$L = \mu\pi a^2 l n^2$$

[34]



$$\vec{E}_{emf} = \vec{v} \times \vec{B} = -r\omega B \cos\theta$$

$$\vec{J} = \sigma \vec{E} \quad (*)$$

$$d\vec{I} = \vec{J}\, dV = \vec{J}\, r\, dr\, d\theta$$

$$d\vec{N} = \vec{r} \times (d\vec{I} \times \vec{B}) = \vec{r} \times (\vec{J} \times \vec{B})\, r\, dr\, d\theta$$

$$= -\sigma r^3 \omega B^2 \cos^2\theta\, dr\, d\theta$$

$$N = -\frac{\pi}{4} \sigma B^2 \omega a^4$$

$$I = \int \rho r^2\, dV = \frac{\pi}{2} \rho a^4$$

$$I \frac{d\omega}{dt} = N \rightarrow \frac{d\omega}{dt} = -\frac{\sigma}{\rho} B^2 \omega$$

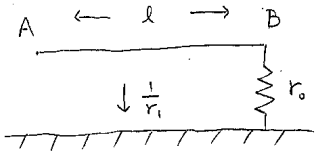
$$\omega = \omega_0 e^{-\frac{\sigma}{\rho} B^2 t}$$

[35]

$j = \sigma E$

$$I = \int j\,dS = \sigma \underbrace{\int E \cdot dS}_{\frac{Q}{\epsilon}} = \underbrace{\frac{\sigma}{\epsilon} CV}_{\frac{1}{R}} \qquad \therefore \quad RC = \frac{\epsilon}{\sigma}$$

[36]



$$V_0 = \int_0^{\ell} r\, I(x)\, dx + r_0\, I(\ell) \qquad \cdots (1)$$

$$V_0 = \int_0^{x} r\, I(x')\, dx' - r_1 \frac{dI}{dx} \qquad \cdots (2)$$

[37]

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^3} \vec{r}$$

$$\vec{B} = \nabla \frac{1}{4\pi} \cdot \frac{\vec{\mu} \cdot \vec{r}}{r^3} = \frac{1}{4\pi} \left(\frac{\vec{\mu}}{r^3} - \frac{3(\vec{\mu} \cdot \vec{r})\vec{r}}{r^5} \right)$$

$$m \ddot{\vec{r}} = q (\vec{E} + \dot{\vec{r}} \times \vec{B})$$

$$r = \left(\frac{3M^2 \hbar^2 a}{2\pi \epsilon_0 m c^4} \left| \frac{q}{Q} \right| \right)^{\frac{1}{3}}$$

$$\omega = \frac{c^2 Q}{3M\hbar^2 a}$$

[39]

$$\vec{M} = \frac{1}{2}\int \vec{r}\times\vec{j}(\vec{r})\,dS = \frac{1}{2}\int a\cdot\sigma a\sin\theta\cdot\omega\cdot(\sin\theta)\left(a^2\sin\theta\,d\theta\,d\varphi\right)$$

$$= \pi\sigma\omega a^4\int_0^{\pi}\sin^3\theta\,d\theta = \frac{4}{3}\pi\sigma\omega a^4 = \frac{1}{3}Q\omega a^2$$

$$B = \nabla\frac{\mu_0}{4\pi}\cdot\frac{\vec{M}\cdot\vec{r}}{r^3} = \frac{\mu_0}{4\pi}\left(\frac{\vec{M}}{r^3} - \frac{3(\vec{M}\cdot\vec{r})\cdot\vec{r}}{r^5}\right) \qquad r \gg a$$

(実は $r\sim a$ でも o.k)

[40] [23] 와 같다.

[41] $\vec{B} = \nabla \times \vec{A} = -\hat{r}\frac{\partial A_\phi}{\partial z} + \hat{z}\frac{1}{r}\frac{\partial}{\partial r}(rA_\phi)$

$d\vec{r} \parallel \vec{B} \rightarrow d\vec{r} \times \vec{B} = 0$

$$\begin{pmatrix} dr \\ d\phi \\ dz \end{pmatrix} \times \begin{pmatrix} B_r \\ 0 \\ B_z \end{pmatrix} = \begin{pmatrix} -B_z\, d\phi \\ B_z\, dr - B_r\, dz \\ -B_r\, d\phi \end{pmatrix} = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}(rA)\,dr + \frac{\partial A}{\partial z}\,dz = 0$$

$\leftarrow \quad \frac{\partial}{\partial r}(rA)\,dr + \frac{\partial}{\partial z}(rA)\,dz = 0 \qquad \therefore\ rA = \text{const}$

[42]